

INT8 MatVec RTL optimization on VTR

Project-contract paired VTR estimate · open 45 nm FPGA proxy

A range-correct balanced adder tree improves the paired area-delay-power estimate by 8.3230%

Prepared by Göther Labs

Measurement completed: 16 August 2026

1. Executive result

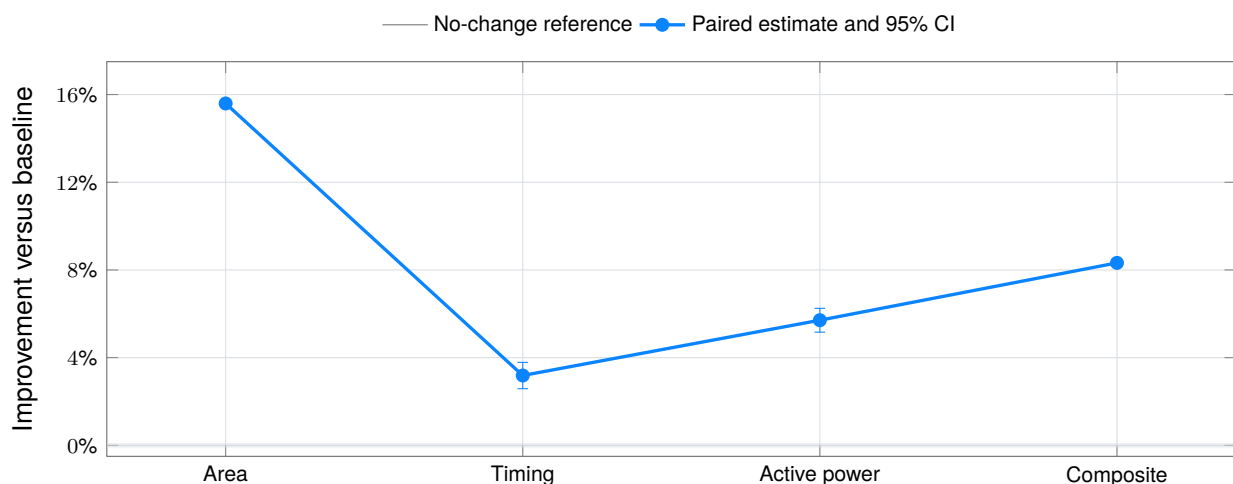
FROZEN BASELINE VERSUS OPTIMIZED RTL

The optimized RTL improves the equal-weight area-delay-active-power estimator by 8.3230% (95% CI: 8.1997% to 8.4462%) without changing the signed INT8 inputs, four signed INT32 outputs or exact matrix-vector arithmetic.

Primary metric	Baseline aggregate	Optimized aggregate	Paired improvement	95% CI / state
Total area	33,892,339 MWTa	28,607,064 MWTa	15.5943%	15.5355% to 15.6531% · better
Critical path	11.1642 ns	10.8085 ns	3.1868%	2.5843% to 3.7855% · better
Active total power	22.4567 mW	21.1749 mW	5.7081%	5.1622% to 6.2509% · better
Composite estimator	1.000000	0.916770	8.3230%	8.1997% to 8.4462% · better

Paired primary PPA improvement

64 fixed held-out certification pairs · lower is better · bars are two-sided 95% confidence intervals



The aggregate values are geometric means across the fixed held-out sample. The reported composite value 0.916770 is the paired geometric-mean estimator; the median of the 64 per-pair composite ratios is 0.916770. The estimator and its paired interval define acceptance.

2. Module and benchmark

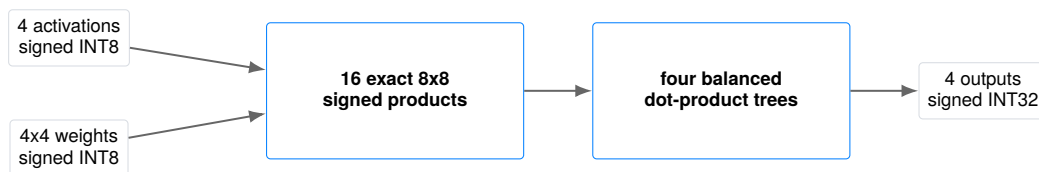
WHAT IS BEING OPTIMIZED

The evaluated module is an owned, Apache-2.0-licensed, synthesizable SystemVerilog implementation of a signed 4x4 matrix-vector product. Four signed 8-bit activations and sixteen signed 8-bit weights produce four exact signed 32-bit dot products:

$$y_i = \sum_{j=0}^3 w_{ij}x_j, \quad i \in \{0, 1, 2, 3\}.$$

Quantized linear, projection and convolution operators repeatedly use this arithmetic primitive. The case deliberately isolates the compute kernel; it is not a complete layer, memory system, neural network, accelerator platform or deployment.

```
module int8_matvec_4x4 (
  input logic signed [7:0] x0, x1, x2, x3,
  input logic signed [7:0] w00, w01, ... , w32, w33,
  output logic signed [31:0] y0, y1, y2, y3
);
```



Only `int8_matvec_4x4.sv` is editable. Port names, signedness, widths, combinational behaviour and overflow semantics are frozen. Every result is compared with the checked-in owned baseline.

The DUT itself is combinational. A trusted, identical measurement wrapper registers only the four 32-bit outputs to provide a stable input-to-output-register timing boundary. Its 72 registers are fixed evaluation infrastructure, not candidate state. The homogeneous target has LUT6 logic and flip-flops but no commercial DSP slices, BRAMs or ASIC MAC cells; the sixteen signed multipliers therefore map into LUT fabric.

Artifact	SHA-256
Frozen baseline SystemVerilog	c2472728e2d52b29ad451f5c9779ebb749c318e9be048d03e6a6555c735eda53
Optimized SystemVerilog	25327189f78f238e909fa5940aff9e2b000547336cdc614ee9ea9c4633aa11f

3. Evaluation contract

FROZEN BEFORE CERTIFICATION

Dimension	Frozen choice
Editable artifact	<code>int8_matvec_4x4.sv</code> only; submissions frozen by SHA-256.
Functional behaviour	Exact ports, signed INT8 operands, signed INT32 outputs and combinational dot-product result.
Arithmetic semantics	Each 8x8 product is exact signed 16-bit; each four-term sum is exact and sign-extended to 32 bits.
Golden simulation	151 deterministic extreme, lane, row and seeded-random cases.
Formal semantics	Exhaustive two-state combinational equivalence for every 160-bit input assignment.
Physical flow	VTR/VPR commit <code>95f5c6de...</code> ; pinned Linux/amd64 image; runtime network disabled.
Architecture	<code>k6_N10_I40_Fi6_L4_frac0_ff1_45nm.xml</code> ; homogeneous LUT6 target, no DSP or BRAM model.
Timing boundary	Fixed trusted wrapper registers the four outputs; identical for baseline and accepted RTL.
Power	VTR PTM45, 0.9 V, 85 °C; fixed active and idle activity profiles.
Search sample	5 paired placements; provisional ranking only.
Certification sample	64 disjoint held-out pairs; fixed sample with no extension.
Primary score	Equal-weight paired area, delay and active-total-power geometric mean.
Validity envelope	Fail closed on missing metrics, route failure, oversized multiplier structure or declared resource limit.



Exact decision rule. All correctness and evidence-integrity gates must pass. The composite one-sided 95% upper confidence bound must be below 1.0, and no primary metric may have a one-sided 95% lower bound above 1.0. Otherwise the result is inconclusive or regressed.

4. Verification before PPA

FAIL-CLOSED EVIDENCE FOR THE OPTIMIZED HASH

Gate	Published evidence	Result
Structural policy	Exact interface, patch and synthesizable SystemVerilog identities	PASS
Golden simulation	151 deterministic signed, extreme, lane, row and random cases	PASS
Combinational formal	All four outputs for every 160-bit binary input assignment	PASS
Physical validity	Width-sensitive mapping, resource envelope and complete metric extraction	PASS
Separate replay	Baseline and optimized records reproduce exactly; not third-party work	PASS

The mathematical reference evaluates signed integer dot products independently of the RTL expression shape. It covers zeros, unit impulses, alternating signs, -128 , 127 , maximum positive and negative accumulation, individual rows and lanes, plus deterministic seeded vectors. A mismatch stops evaluation before synthesis-based scoring.

The formal proof is unbounded because the design is combinational: it compares every output bit for every possible assignment of the twenty signed INT8 inputs. This is 2^{160} input assignments represented symbolically, not a claim that they were enumerated one by one. Inconclusive or timed-out proof is invalid.

Implementation validity rejects missing or non-finite metrics, failed routing, inconsistent area/frequency/power identities, multiplier hard blocks, multiplier operands above the declared width and excessive logic resources. The earlier syntax-level AST count is not an authoritative PPA metric.

```
python3 verify.py
```

Evidence boundary. The compact check establishes certificate consistency and exact published identities. With the separately downloaded raw archive, the same verifier verifies every member, checks all four functional and formal pass logs, and re-extracts all 256 blinded post-route PPA rows. It does not rerun Verilator, Yosys or VTR.

5. Exact baseline-to-optimized diff

ALL CHANGED LINES FROM INT8_MATVEC_4X4.SV

The patch below is generated directly from the two checked-in RTL files. No testbench, reference model, architecture, activity profile, metric parser or tool flag is editable by the optimization candidate.

```
--- rtl/baseline/int8_matvec_4x4.sv
+++ rtl/optimized/int8_matvec_4x4.sv
@@ -32,10 +32,11 @@
+ logic signed [15:0] p10, p11, p12, p13;
+ logic signed [15:0] p20, p21, p22, p23;
+ logic signed [15:0] p30, p31, p32, p33;
- logic signed [31:0] e00, e01, e02, e03;
- logic signed [31:0] e10, e11, e12, e13;
- logic signed [31:0] e20, e21, e22, e23;
+ logic signed [31:0] e30, e31, e32, e33;
+ logic signed [16:0] s00, s01;
+ logic signed [16:0] s10, s11;
+ logic signed [16:0] s20, s21;
+ logic signed [16:0] s30, s31;
+ logic signed [17:0] d0, d1, d2, d3;

always_comb begin
    p00 = w00 * x0;
@@ -55,26 +56,23 @@
    p32 = w32 * x2;
    p33 = w33 * x3;

-    e00 = {{16{p00[15]}}, p00};
-    e01 = {{16{p01[15]}}, p01};
-    e02 = {{16{p02[15]}}, p02};
-    e03 = {{16{p03[15]}}, p03};
-    e10 = {{16{p10[15]}}, p10};
-    e11 = {{16{p11[15]}}, p11};
-    e12 = {{16{p12[15]}}, p12};
-    e13 = {{16{p13[15]}}, p13};
-    e20 = {{16{p20[15]}}, p20};
-    e21 = {{16{p21[15]}}, p21};
-    e22 = {{16{p22[15]}}, p22};
-    e23 = {{16{p23[15]}}, p23};
-    e30 = {{16{p30[15]}}, p30};
-    e31 = {{16{p31[15]}}, p31};
-    e32 = {{16{p32[15]}}, p32};
-    e33 = {{16{p33[15]}}, p33};
+    s00 = {p00[15], p00} + {p01[15], p01};
+    s01 = {p02[15], p02} + {p03[15], p03};
+    s10 = {p10[15], p10} + {p11[15], p11};
+    s11 = {p12[15], p12} + {p13[15], p13};
+    s20 = {p20[15], p20} + {p21[15], p21};
+    s21 = {p22[15], p22} + {p23[15], p23};
+    s30 = {p30[15], p30} + {p31[15], p31};
+    s31 = {p32[15], p32} + {p33[15], p33};

-    y0 = e00 + e01 + e02 + e03;
-    y1 = e10 + e11 + e12 + e13;
-    y2 = e20 + e21 + e22 + e23;
-    y3 = e30 + e31 + e32 + e33;
+    d0 = {s00[16], s00} + {s01[16], s01};
+    d1 = {s10[16], s10} + {s11[16], s11};
+    d2 = {s20[16], s20} + {s21[16], s21};
+    d3 = {s30[16], s30} + {s31[16], s31};
+
+    y0 = {{14{d0[17]}}, d0};
+    y1 = {{14{d1[17]}}, d1};
+    y2 = {{14{d2[17]}}, d2};
+    y3 = {{14{d3[17]}}, d3};
end
endmodule
```

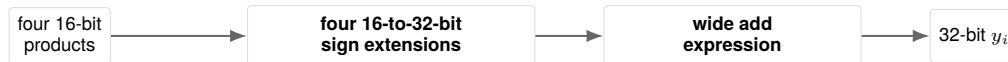
Attribution boundary. The measured PPA result belongs to one complete accepted patch. The following pages are three overlapping structural views of that same patch, not three independently measured rewrites; no exact fraction of the gain is assigned to one statement.

6. Structural view 1: remove full-width extension bank

ONE ACCEPTED PATCH · SOURCE: INT8_MATVEC_4X4.SV:35–43 AND 59–74

The baseline materializes sixteen signed 32-bit aliases before any addition. Every product is already exact in signed 16-bit form, so widening each term directly to the 32-bit output width is unnecessary.

Frozen baseline



Optimized RTL



For signed INT8 values $x, w \in [-128, 127]$, each product lies in

$$wx \in [-16,256, 16,384].$$

A pair sum therefore lies in $[-32,512, 32,768]$ and needs 17 signed bits. The complete four-term dot product lies in $[-65,024, 65,536]$ and needs 18 signed bits. No 32-bit intermediate is required before the interface boundary.

Formal result. Exhaustive combinational equivalence proves that removing the extension bank preserves every output bit for every binary input assignment.

7. Structural view 2: balanced range-correct accumulation

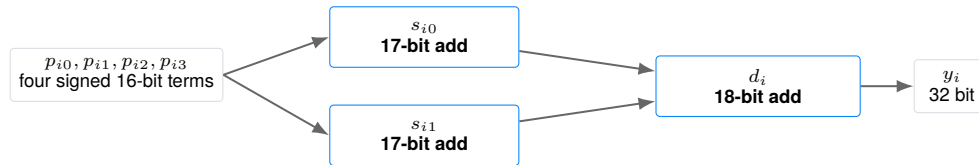
ONE ACCEPTED PATCH · SOURCE: INT8_MATVEC_4X4.SV:59–70

The baseline presents four already widened terms as one 32-bit chained expression and leaves its mapped association to synthesis. The optimized RTL makes both the intended topology and the minimum safe widths explicit.

Frozen baseline · wide source expression



Optimized RTL · explicit balanced tree



For one row, the implementation is

$$\begin{aligned}
 s_{i0} &= \text{sext}_{17}(p_{i0}) + \text{sext}_{17}(p_{i1}), \\
 s_{i1} &= \text{sext}_{17}(p_{i2}) + \text{sext}_{17}(p_{i3}), \\
 d_i &= \text{sext}_{18}(s_{i0}) + \text{sext}_{18}(s_{i1}).
 \end{aligned}$$

Integer addition is associative because every intermediate width covers its full mathematical range. The two-level topology avoids relying on tool interpretation of one wide chained expression and exposes width-sensitive mapping to synthesis.

Scope. This transformation is specific to four exact INT8 products and a 32-bit output contract. A different lane count, operand width, saturation rule or accumulation precision requires a new range proof and evaluation contract.

8. Structural view 3: explicit signed output boundary

ONE ACCEPTED PATCH · SOURCE: INT8_MATVEC_4X4.SV:72–75

The baseline moves every 16-bit product into a 32-bit representation before addition. The optimized path remains range-sized through the complete sum and crosses the frozen 32-bit interface only once. No truncation, wraparound or saturation occurs.

Frozen baseline · widen before accumulation



Optimized RTL · widen once at output



For every representable result d_i ,

$$\text{value}_{32}(\{\{14\{d_i[17]\}\}, d_i\}) = \text{value}_{18}(d_i) = \sum_{j=0}^3 w_{ij}x_j.$$

The output assignment changes neither combinational latency semantics nor the observable 32-bit value. It only prevents a wide internal representation from propagating backwards through the arithmetic cone.

Correctness result. All 151 deterministic simulations and exhaustive formal equivalence agree on all four signed INT32 outputs. The accepted PPA measurement is therefore downstream of, and conditional on, exact arithmetic correctness.

9. PPA environment and statistics

OPEN 45 NM FPGA PROXY

The target matches the SHA-1 methodology case: VTR's homogeneous `k6_N10_I40_Fi6_L4_frac0_ff1_45nm` architecture, with six-input LUTs clustered ten per logic block and the architecture's routing and timing model. Power uses PTM45 properties at 0.9 V and 85 °C. MWTa is minimum-width-transistor area, not square-micrometre physical area.

Two 2,048-cycle activity profiles are frozen: an active deterministic signed-INT8 workload containing extremes and a fixed xorshift64 sequence, and an idle all-zero data profile with the clock active. Active and idle power analyses reuse the same placement and route for a given design and blinded pair identity.

The primary per-pair score is

$$r_p = \sqrt[3]{r_{A,p} r_{D,p} r_{P,p}}, \quad \hat{r} = \exp\left(\frac{1}{64} \sum_{p \in P} \log r_p\right),$$

where A is total MWTa, D is post-route critical-path delay and P is active total power. $r_{\max}$ is reported as the reciprocal of routed delay and is not a separate score term.

Two-sided 95% Student- t intervals use 63 degrees of freedom on paired log ratios. The one-sided bounds in the decision rule use the same predeclared fixed sample. These intervals quantify variation across the selected implementation pairs; they do not cover process, voltage, temperature, workload or tool-version uncertainty.

Pair policy. Search uses only 5 paired placements. Certification uses 64 disjoint held-out pairs after the optimized RTL hash is frozen. The public certificate replaces reserved placement-seed identities with stable pair labels; the optimizer and candidate do not see the certification pool.

10. Every paired certification result

PRIMARY PPA DISTRIBUTIONS



FIGURE 2 · OVERLAID BASELINE AND OPTIMIZED PAIR CLOUDS

Gray and blue points are absolute results normalized to the same baseline geometric aggregate. The same blinded pair receives the same vertical jitter. Black points and segments show the paired estimator and interval that determine confidence.

Metric	Wins	Ties	Losses
Area	64	0	0
Timing	54	0	10
Active power	64	0	0
Composite	64	0	0

11. Secondary metrics and evidence identity

MEASURED CONSEQUENCES AND PUBLIC AUDIT TRAIL

Metric	Baseline aggregate	Optimized aggregate	Interpretation
Fmax	89.5717 MHz	92.5202 MHz	3.2918% increase; reciprocal of routed delay
Packed CLBs	368	304	17.3913% fewer in every pair
Logic elements	2586	2467	4.6017% fewer in every pair
Fixed wrapper registers	72	72	Unchanged; not candidate datapath state
Idle total power	9.1024 mW	8.1258 mW	10.7284% paired geometric improvement; excluded from score

The accepted design uses no multiplier hard blocks because this academic architecture provides none: all multiply logic is LUT-mapped. It remains within the frozen resource envelope. Dynamic and static components are retained per pair, but only active total power enters the primary score. None of these fields is inferred from an RTL node count.

Public evidence object	SHA-256 / identity
Compact certification record	e77ed39c79559af8cbbd2efeaf4fb67045740684ac4fc35699213c3cc1858906
Functional test-set identity	87abb71f1542981919fb62e299e0c19ff4cfacae20f41d50d0a41756d345c513
VTR architecture identity	262792caef81931ae09b77d252158bde03c78954175d4b761e6d437317575307
PTM technology identity	3080dea13bf7134109f8a83c79426ec0965e07639a61866fe9ae4bd05b5227cb
Optimized primary record	5f42601df61768be36b23594205c8d61d138bb4ed760443af0bbef0b43948a7d
Optimized replay record	441b9da134214162176d16a79e740dc50c1cdb1c9a244e6a1fa640b906304a6d
Full raw-evidence archive	int8-matvec-vtr45-full-evidence-v1.tar.gz
Full archive SHA-256	ecfa2f611438cadeb21af8de5d6d6e5f4fecc3546618a7eab6b6f54f8cb57f04

The certificate also binds distinct primary and replay records for the baseline. The release archive supplies the underlying functional, formal and routed VTR outputs for both RTL variants and both evidence roles while keeping reserved placement identities blinded.

12. Reproduction and verifier trust model

WHAT EACH PUBLIC COMMAND ESTABLISHES

The repository separates four levels of evidence:

1. **Git identity:** checksums cover the exact compact package and the commit records its history.
2. **Compact consistency:** `python3 verify.py` validates strict schemas, RTL hashes, the exact patch, all 64 paired rows, derived metrics, formulas, confidence intervals, resource limits and acceptance rules.
3. **Raw provenance replay:** `python3 verify.py --evidence-archive ...` verifies the hash-addressed archive, functional and formal pass logs, four 64-pair evidence legs and PPA values re-extracted from VTR timing and power reports.
4. **Separate project replay:** the primary and replay legs are separate complete tool runs; exact agreement is checked through the public raw evidence. This does not establish organisational independence.

```
git clone https://github.com/juan-fernandez-gotherlabs/\
rtl-optimization-case-study.git
cd rtl-optimization-case-study/cases/int8-matvec

python3 verify.py
python3 verify.py --evidence-archive \
    int8-matvec-vtr45-full-evidence-v1.tar.gz
python3 -m unittest discover -s tests -v
```

The verifier uses explicit checks rather than Python `assert`, requires exact checksum coverage, rejects duplicate or unexpected schema fields, non-finite measurements, altered RTL, authority drift, metric substitutions, score substitutions and any leaked held-out seed identity. Its adversarial unit suite exercises these cases in CI.

A fresh EDA rerun requires Linux/amd64, the pinned VTR commit, exact architecture and PTM files, fixed activity profiles and the published evaluation contract. The customer-facing package intentionally excludes the optimization implementation and operational execution environment because neither is needed to audit the delivered before/after result.

Trust boundary. The Git commit and release identify the published package. The raw archive permits provenance and metric re-extraction comparable to SHA-1, but remains recorded evidence: it is not a fresh third-party tool execution and cannot establish physical-chip behaviour. “Certification” here means project-contract acceptance, not accredited assurance.

13. Limits and customer transfer

WHAT THE RESULT DOES AND DOES NOT ESTABLISH

This report does not provide:

- ASIC synthesis, place-and-route or signoff;
- commercial-FPGA implementation data, DSP/BRAM mapping or Vivado results;
- extracted parasitics, PVT corners, clock-tree signoff, IR drop, EM, DRC/LVS, package, yield or production-test analysis;
- board, silicon, measured-power or measured-energy evidence;
- proof that the same structure improves a different matrix size, precision, architecture, technology or workload;
- exact PPA attribution to one line or one structural aspect of the patch;
- a complete neural-network layer, memory hierarchy or deployable AI accelerator.

The 64-pair interval models implementation-seed variability for the fixed VTR contract. It does not model silicon process variation or generalise automatically to a customer technology.

Transfer to a customer pilot. Replace the academic proxy with customer-owned RTL, formal assumptions, libraries, constraints, activity, tools and signoff policy. The transferable capability is a small inspectable source change evaluated by a contract that can justify acceptance to the engineers responsible for the design.

Primary references

- [VTR/VPR documentation](#)
- [VTR power estimation](#)
- [Yosys documentation](#)
- [Public case-study repository](#)
- [Göther Labs](#)

Conclusion. Within the declared VTR 45 nm LUT-only FPGA proxy and exact signed-arithmetic contract, the accepted balanced-tree patch passes the recorded functional and exhaustive formal gates and reduces the paired primary PPA estimator by 8.3230% (95% CI 8.1997% to 8.4462%). The before/after RTL, compact calculations, raw primary runs and separate project replay are publicly verifiable. This remains academic implementation evidence, not a physical measurement or an externally audited result.